REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 3

Orbit Transfers and Delta-v

Changing Orbits on Purpose: Hohmann, Bi-elliptic, Plane Changes and Low-thrust

Prepared for	Cao Khanh Tan
Module	3, continuing the Core Training Phase
Builds on	Sessions 1 and 2: vis-viva, the GTO you sized, J2 station-keeping
Tools	poliastro Maneuver, GMAT targeting, STK Astrogator

SESSION 3

Orbit Transfers and Delta-v

Session 2 spent delta-v to fight perturbations and hold an orbit in place. This module spends it on purpose, to move a spacecraft from one orbit to another. Every manoeuvre is a velocity change, delta-v, and the whole subject is really one question asked in many forms: what is the cheapest set of burns that gets me from orbit A to orbit B? The tool for answering it is the vis-viva equation you have used since Session 1. The best news is that you have already done most of the first example: the GTO you sized in Session 1 is exactly the transfer ellipse we will now turn into a full Earth-to-GEO mission.

Learning objectives

- Define delta-v and connect it to propellant through the rocket equation.
- Derive and compute the two-burn Hohmann transfer between coplanar circular orbits.
- Know when a three-burn bi-elliptic transfer beats Hohmann, and why.
- Compute the cost of a plane change and explain why it is done at apoapsis.
- Combine an altitude change and a plane change into one efficient burn.
- Contrast impulsive and continuous low-thrust transfers on delta-v, propellant and time.

> **The one idea behind all of it**
> A burn changes your speed at a point; vis-viva tells you what speed each orbit needs at that point; the difference is your delta-v. Master that loop and every transfer in this module is the same calculation applied to different geometry.

PART 1

Delta-v and the Rocket Equation

We model chemical burns as impulsive: instantaneous changes in velocity at a single point, since a burn lasts minutes against an orbit of hours. The magnitude of that change is the delta-v of the manoeuvre, and it is the true currency of spaceflight because it maps directly to propellant through the Tsiolkovsky rocket equation:

$$\Delta v = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

Here Isp is the specific impulse of the engine and g0 is standard gravity. Rearranged, it tells you the propellant mass fraction a given delta-v demands:

$$\frac{m_p}{m_0} = 1 - e^{-\Delta v/(I_{sp} g_0)}$$

The relationship is exponential, which is the tyranny of the rocket equation: a 3.8 km/s transfer on a chemical stage (Isp about 320 s) burns roughly 70 percent of the total mass as propellant. The same delta-v on an electric thruster (Isp about 3000 s) needs only about 12 percent, which is exactly why low-thrust is attractive, as we will see in Part 5. Minimising delta-v is therefore not academic: it decides whether a mission closes at all.

Recall the tool
Every delta-v below is a difference of two speeds computed from vis-viva, v squared equals mu times (2/r minus 1/a). Keep it in front of you.

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$$

PART 2

The Hohmann Transfer

The Hohmann transfer is the minimum-energy way to move between two coplanar circular orbits using two burns. It rides a single ellipse that is tangent to the inner orbit at its perigee and to the outer orbit at its apogee, so both burns are pure prograde or retrograde, with no wasted sideways component. The transfer ellipse has semi-major axis:

$$a_t = \frac{r_1 + r_2}{2}$$

The first burn raises perigee speed from the inner circular speed to the transfer perigee speed; the second, at apogee, raises the transfer apogee speed up to the outer circular speed:

$$\Delta v_1 = v_p - v_{c1}, \qquad \Delta v_2 = v_{c2} - v_a$$

Written out with vis-viva, the two burns have the closed forms:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1\right)$$

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right)$$

and the trip takes exactly half of the transfer ellipse period:

$$t_{\text{transfer}} = \pi\sqrt{\frac{a_t^3}{\mu}}$$

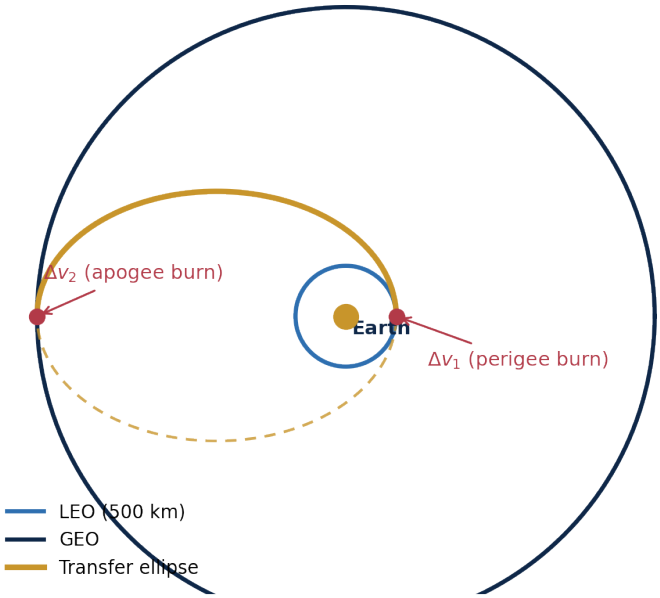


Figure 1. The Hohmann transfer from a 500 km LEO to GEO. This is the very ellipse you sized in Session 1: perigee at the LEO radius, apogee at the GEO radius. The perigee burn injects onto the ellipse; the apogee burn circularises at GEO.

Worked example: LEO to GEO (your GTO, finished)
With r1 = 6,878 km and r2 = 42,164 km: the inner circular speed is 7.61 km/s and the transfer perigee speed is 9.98 km/s, so the first burn is delta-v1 = 2.37 km/s. At apogee the transfer speed is 1.63 km/s and GEO needs 3.07 km/s, so delta-v2 = 1.45 km/s. The total is about 3.82 km/s, and the flight takes half the GTO period, about 5.3 hours. Note the two speeds feeding this: 9.98 and 1.63 km/s, the corrected values from your Session 1 Task 3.

PART 3

Bi-elliptic Transfers

Hohmann is optimal only up to a point. For very large radius ratios a three-burn bi-elliptic transfer can cost less total delta-v. It flings the spacecraft far beyond the target to a high intermediate apogee r_b, raises perigee there cheaply (because the vehicle is moving slowly), then drops back to circularise at the target:

$$\Delta v_{\text{bi}} = \Delta v_1 + \Delta v_2 + \Delta v_3 \quad \text{(three burns via } r_b > r_2\text{)}$$

The trade is time for fuel: bi-elliptic transfers take far longer, sometimes many times the Hohmann duration. The crossover depends on the radius ratio:

- For r2 / r1 below about 11.94, Hohmann always wins.
- For r2 / r1 above about 15.58, bi-elliptic always wins (with a high enough intermediate apogee).
- Between those two values it depends on the intermediate radius r_b you choose.

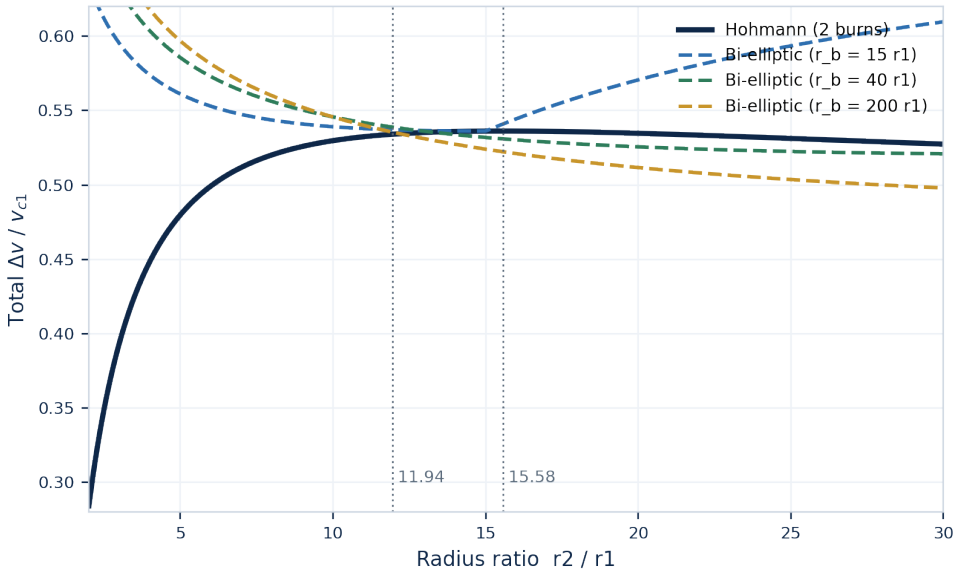


Figure 2. Total delta-v (normalised by the inner circular speed) versus radius ratio. The Hohmann curve actually peaks near a ratio of 15.6 and then decreases, while bi-elliptic transfers with a higher intermediate apogee keep improving. The two dotted lines mark the classic 11.94 and 15.58 thresholds.

PART 4

Plane Changes and Combined Burns

Everything so far kept the orbit in one plane. Changing the orbital plane, for example rotating inclination to reach an equatorial GEO from an inclined launch, is one of the most expensive things you can do. A pure rotation of the velocity vector by an angle delta i at speed v costs:

$$\Delta v_{\text{plane}} = 2\, v \sin\left(\frac{\Delta i}{2}\right)$$

Because the cost scales with the speed v at which you turn, you should always change plane where you are moving slowest, which is at apoapsis. Turning at GEO apogee costs a fraction of what the same rotation would cost down in LEO. Better still, you rarely do a plane change on its own: you fold it into the circularisation burn so a single manoeuvre changes both speed and direction at once. The combined cost follows the law of cosines:

$$\Delta v = \sqrt{v_1^2 + v_2^2 - 2\, v_1 v_2 \cos \Delta i}$$

Worked example: removing 28.5 degrees at GEO
A launch from Cape Canaveral reaches GTO at about 28.5 degrees inclination and must end up equatorial. Doing the plane change separately at GEO speed costs 2 x 3.07 x sin(14.25 deg) = 1.51 km/s on its own. But combining it with the apogee circularisation burn (v_a = 1.63, v_c = 3.07 km/s) costs 1.82 km/s for the whole thing, versus 1.45 km/s for circularisation alone. The entire 28.5 degree rotation is bought for only about 0.37 km/s extra. That is why GEO insertion is always a combined apogee burn.

PART 5

Low-thrust Transfers

Electric thrusters produce tiny forces continuously for months rather than large impulses in minutes. Instead of two tangent burns, the spacecraft spirals outward through hundreds of revolutions. For a coplanar circle-to-circle spiral, the Edelbaum result is strikingly simple, just the difference of the two circular speeds:

$$\Delta v_{\mathrm{LT}} = \left| v_{c1} - v_{c2} \right| \quad \text{(coplanar, continuous thrust)}$$

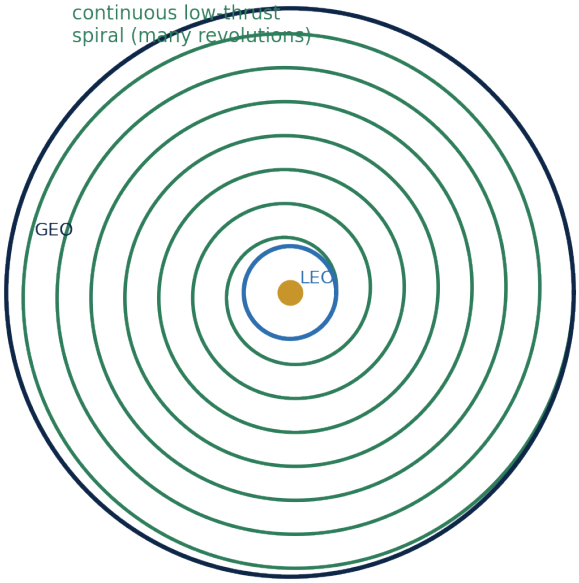


Figure 3. A continuous low-thrust spiral from LEO to GEO. There are no distinct burns; thrust is applied along the velocity over many revolutions.

For LEO to GEO this gives about 4.54 km/s, which is more delta-v than Hohmann (3.82 km/s). Yet because the specific impulse of electric propulsion is roughly ten times higher, the propellant mass is dramatically lower: the rocket equation rewards the high Isp far more than it punishes the extra delta-v. The price is time (months instead of hours) and many passes through the radiation belts. Choosing between chemical and electric is therefore a mission-level trade between propellant mass and schedule.

Computing Transfers

The Hohmann calculation is a few lines of vis-viva. The snippet below reproduces the LEO-to-GEO worked example from scratch, and the comment shows the one-line poliastro equivalent.

```
import numpy as np
mu = 398600.4418            # km^3/s^2
r1, r2 = 6378.137+500, 42164.0     # LEO 500 km, GEO

vc  = lambda r: np.sqrt(mu/r)
vis = lambda r,a: np.sqrt(mu*(2/r - 1/a))
a_t = (r1 + r2)/2

dv1 = vis(r1,a_t) - vc(r1)         # perigee burn
dv2 = vc(r2) - vis(r2,a_t)         # apogee burn
t   = np.pi*np.sqrt(a_t**3/mu)     # transfer time
print(dv1, dv2, dv1+dv2, t/3600)   # ~2.37, 1.45, 3.82 km/s, 5.3 h

# poliastro one-liner:
#   from poliastro.maneuver import Maneuver
#   hoh = Maneuver.hohmann(orbit_leo, r2*u.km);  hoh.get_total_cost()
```

Production tools

poliastro Maneuver.hohmann and Maneuver.bielliptic return the burns and total cost directly. GMAT models transfers with its Target sequence and a differential corrector to hit the exact final state; STK does the same through Astrogator mission-control sequences. All of them are solving the vis-viva bookkeeping you just did by hand.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: Orbit Transfers and Delta-v

- **Reading and notes (3 to 4 h).** Explain in your own words why both Hohmann burns are purely tangential, and why a plane change is cheapest at apoapsis. One page.
- **Hohmann by hand and code (4 to 5 h).** Reproduce the LEO-to-GEO transfer (delta-v1, delta-v2, total, time). Then compute the Hohmann transfer from a 400 km LEO to a 20,200 km MEO (a GPS orbit) and report both burns and the transfer time.
- **Bi-elliptic comparison (3 to 4 h).** For a transfer from LEO to a radius ratio of 20, compute the Hohmann total and a bi-elliptic total with r_b = 40 r1. Which wins, and by how much? Comment on the time penalty.
- **Combined burn (2 h).** Repeat the GEO insertion but assume a 51.6 degree launch inclination. How much does the combined apogee burn cost now, and how does that compare to changing plane separately?
- **STK or GMAT (2 h).** Build the LEO-to-GEO Hohmann in Astrogator or a GMAT Target sequence and confirm your hand delta-v to within a few percent. Capture the transfer plot.

Answer key (for your self-check)

Quantity	Expected value
LEO(500)->GEO Hohmann	dv1 = 2.37, dv2 = 1.45, total = 3.82 km/s; ~5.3 h
LEO(400)->MEO(20,200 km)	dv1 about 1.31, dv2 about 1.01 km/s; ~3.1 h
Ratio 20: Hohmann vs bi-elliptic	~4.07 vs ~4.00 km/s (bi-elliptic wins, but far slower)
28.5 deg combined GEO insertion	~1.82 km/s combined vs ~1.45 in-plane only
Rocket eq, 3.82 km/s at Isp 320 s	propellant fraction about 0.70

REFERENCE

One-Page Formula Sheet

Vis-viva (the master tool)	$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$
Transfer ellipse SMA	$a_t = \frac{r_1 + r_2}{2}$
Hohmann first burn	$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1\right)$
Hohmann second burn	$\Delta v_2 = \sqrt{\frac{\mu}{r_2}}\left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}}\right)$
Transfer time	$t_{\text{transfer}} = \pi\sqrt{\frac{a_t^3}{\mu}}$
Bi-elliptic (three burns)	$\Delta v_{\text{bi}} = \Delta v_1 + \Delta v_2 + \Delta v_3$ (three burns via $r_b > r_2$)
Plane change	$\Delta v_{\text{plane}} = 2\,v\sin\left(\frac{\Delta i}{2}\right)$
Combined altitude + plane	$\Delta v = \sqrt{v_1^2 + v_2^2 - 2\,v_1 v_2 \cos\Delta i}$
Low-thrust (Edelbaum, coplanar)	$\Delta v_{\text{LT}} = \lvert v_{c1} - v_{c2} \rvert$ (coplanar, continuous thrust)
Rocket equation	$\Delta v = I_{sp}\, g_0 \ln\frac{m_0}{m_f}$

Key glossary

- **Delta-v:** the magnitude of a commanded velocity change; the currency of every manoeuvre.
- **Impulsive burn:** an idealised instantaneous velocity change (good for chemical engines).
- **Specific impulse (Isp):** engine efficiency in seconds; higher Isp means less propellant per delta-v.
- **Tangential burn:** a burn along the velocity vector, which changes energy most efficiently.
- **GTO:** geostationary transfer orbit, the Hohmann ellipse from LEO to GEO you sized in Session 1.

Going further

For depth: Curtis, Orbital Mechanics for Engineering Students (preliminary orbit design and manoeuvres); Vallado, Fundamentals of Astrodynamics and Applications (maneuvers and mission design); and Prussing and Conway, Orbital Mechanics, for optimal transfers. These are real, widely used references, but please confirm exact editions and chapters yourself, as I cannot verify those details from here.

Coming up in Session 4

Interplanetary transfers and gravity assists: the patched-conic method, hyperbolic escape and arrival, and the Earth-to-Mars transfer named on the programme page. It is the Hohmann idea again, now around the Sun, with planets as moving targets.